Assignment 1 – Figures, illustrations and tables

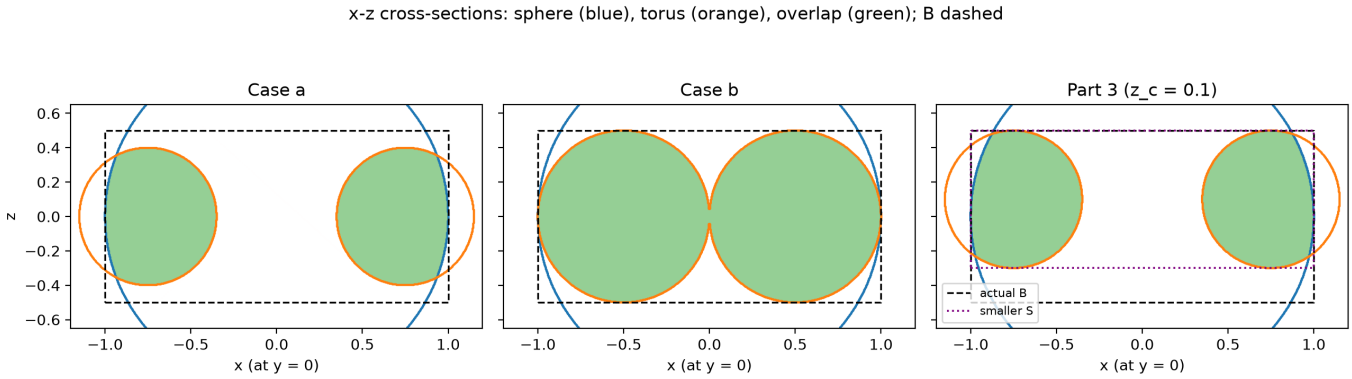


Figure 1

Table 1

Part	Parameters	$\hat{V}$	95% CI ±	Reference	Error
1a	$k = 1, R = 0.75, r = 0.4$	1.8572	0.0124	1.8561	+0.0011
1b	$k = 1, R = 0.5, r = 0.5$	2.4701	0.0120	2.4674	+0.0027
3a	as 1a, centre $(0, 0, 0.1)$	1.8210	0.0123	1.8284	-0.0074

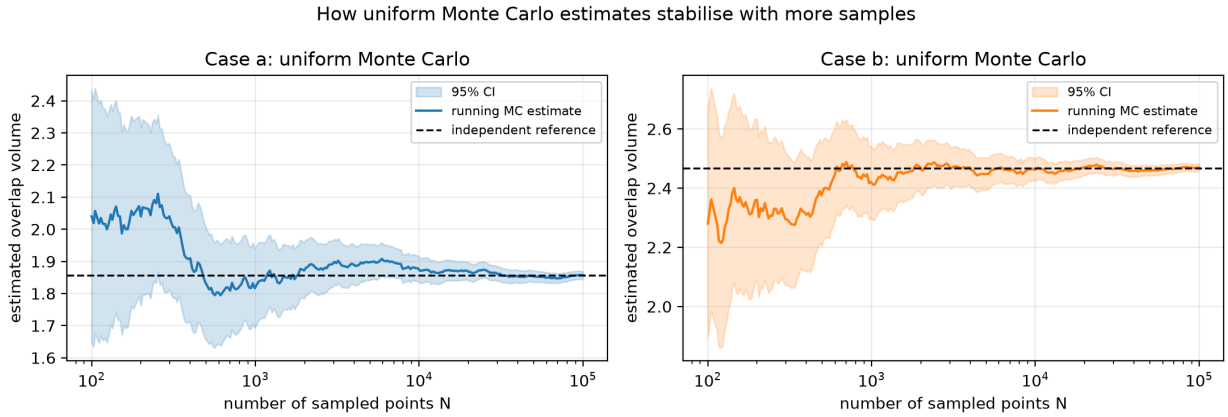


Figure 2

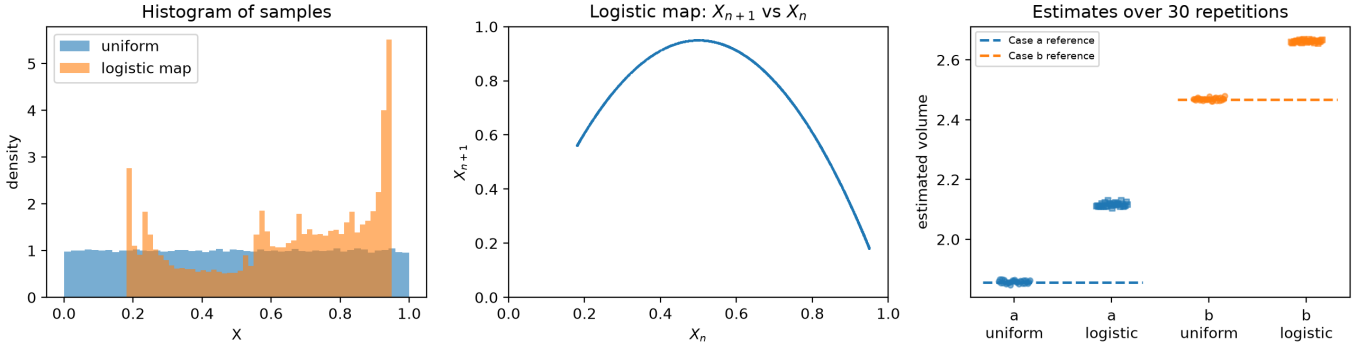


Figure 3

Table 2

Case	Sampler	Mean $\hat{V}$	SD over runs	Mean error	Error/$(s/\sqrt{30})$
a	uniform	1.8578	0.0049	+0.0017	1.9
a	logistic	2.1165	0.0060	+0.2605	237.8
b	uniform	2.4693	0.0044	+0.0019	2.4
b	logistic	2.6627	0.0048	+0.1953	224.1

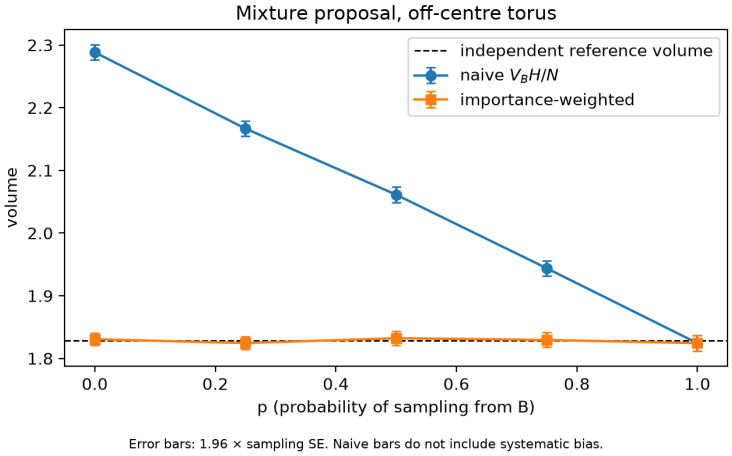


Figure 4

Table 3

p	$\hat{V}_{\text{naive}}$	$E[\hat{V}_{\text{naive}}]$	$\hat{V}_w$	95% CI $\pm$
0.00	2.2883	2.2855	1.8307	0.0098
0.25	2.1666	2.1712	1.8245	0.0104
0.50	2.0612	2.0569	1.8322	0.0110
0.75	1.9438	1.9427	1.8295	0.0117
1.00	1.8244	1.8284	1.8244	0.0123

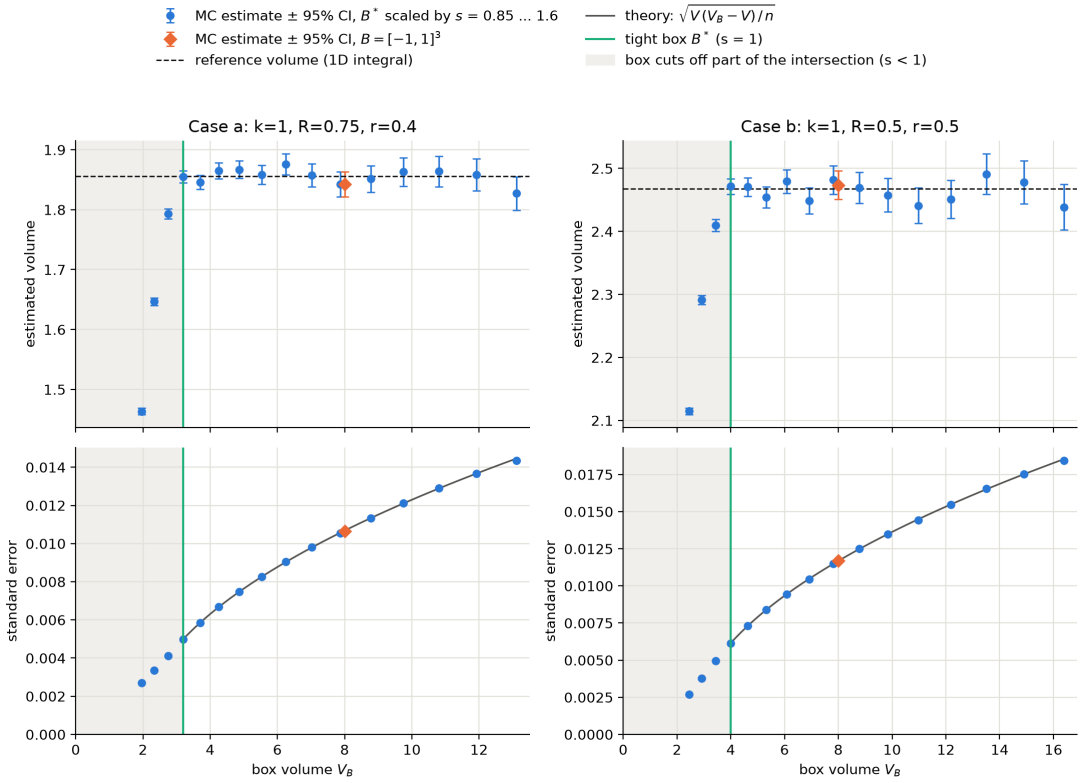

MC estimate ± 95% CI, B^* scaled by s = 0.85 ... 1.6
MC estimate ± 95% CI, $B = [-1, 1]^3$
reference volume (1D integral)
theory: $\sqrt{V(V_B - V)/n}$
tight box B^* (s = 1)
box cuts off part of the intersection (s < 1)
Case a: k=1, R=0.75, r=0.4
Case b: k=1, R=0.5, r=0.5
estimated volume
standard error
box volume V_B

Figure 5